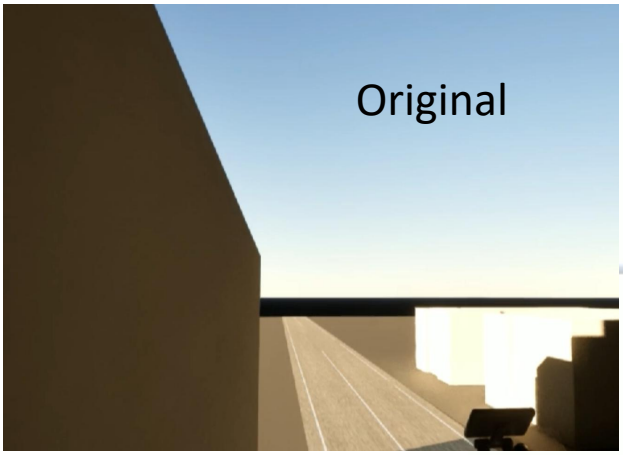
IMGD 4099 Final: Nuclear Explosion Scene

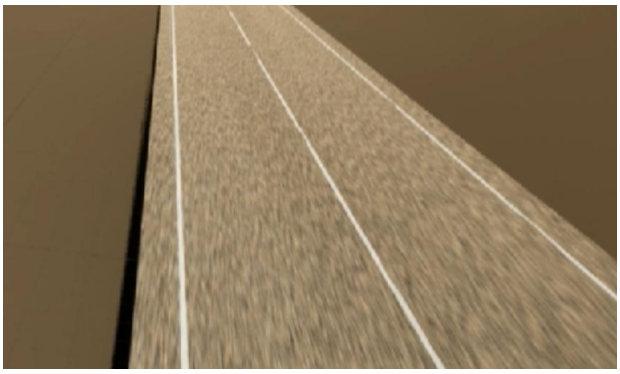
This project is dedicated to simulating the effects of real-time nuclear explosions using Embergen and Unreal Engine. Its core objective is to explore how to integrate various visual effects, such as fluid simulations, lighting, and materials, into a whole scene.

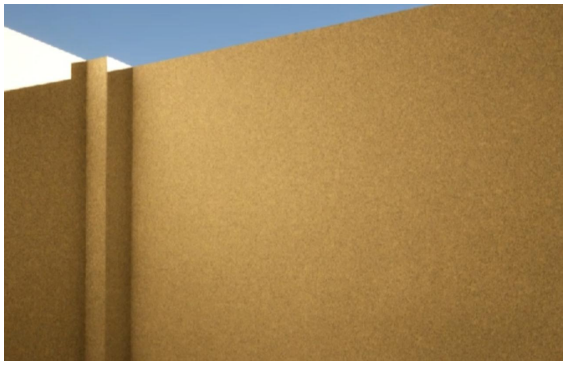
The first step was to create the nuclear mushroom cloud in Embergen. During this process, the project adjusted various parameters to sculpt the explosion's specific characteristics, including smoke emission rates, pressure, fireball position and velocity, and even temperature. The main idea was to craft a "mushroom-like" smoke-and-fireball scene, rather than just a generic fireball or an amorphous cloud of smoke. Subsequently, the Embergen-generated mushroom cloud was imported into Unreal Engine and positioned at an angle that, when viewed from a distance, convincingly simulates a real-time nuclear explosion.

This project simulates lighting from the explosion by dynamically adjusting the scene's exposure. At the time of the explosion, the scene's exposure value increased dramatically, causing the entire frame to flash brightly, then decrease, returning to normal exposure levels. This technique creates the visceral impression that the explosion itself emits light of such overwhelming intensity that it can be felt even by an observer with their back to the blast site. However, a limitation of this effect is that it can only be demonstrated within a live playback environment, as its execution is driven by keyframe animation.

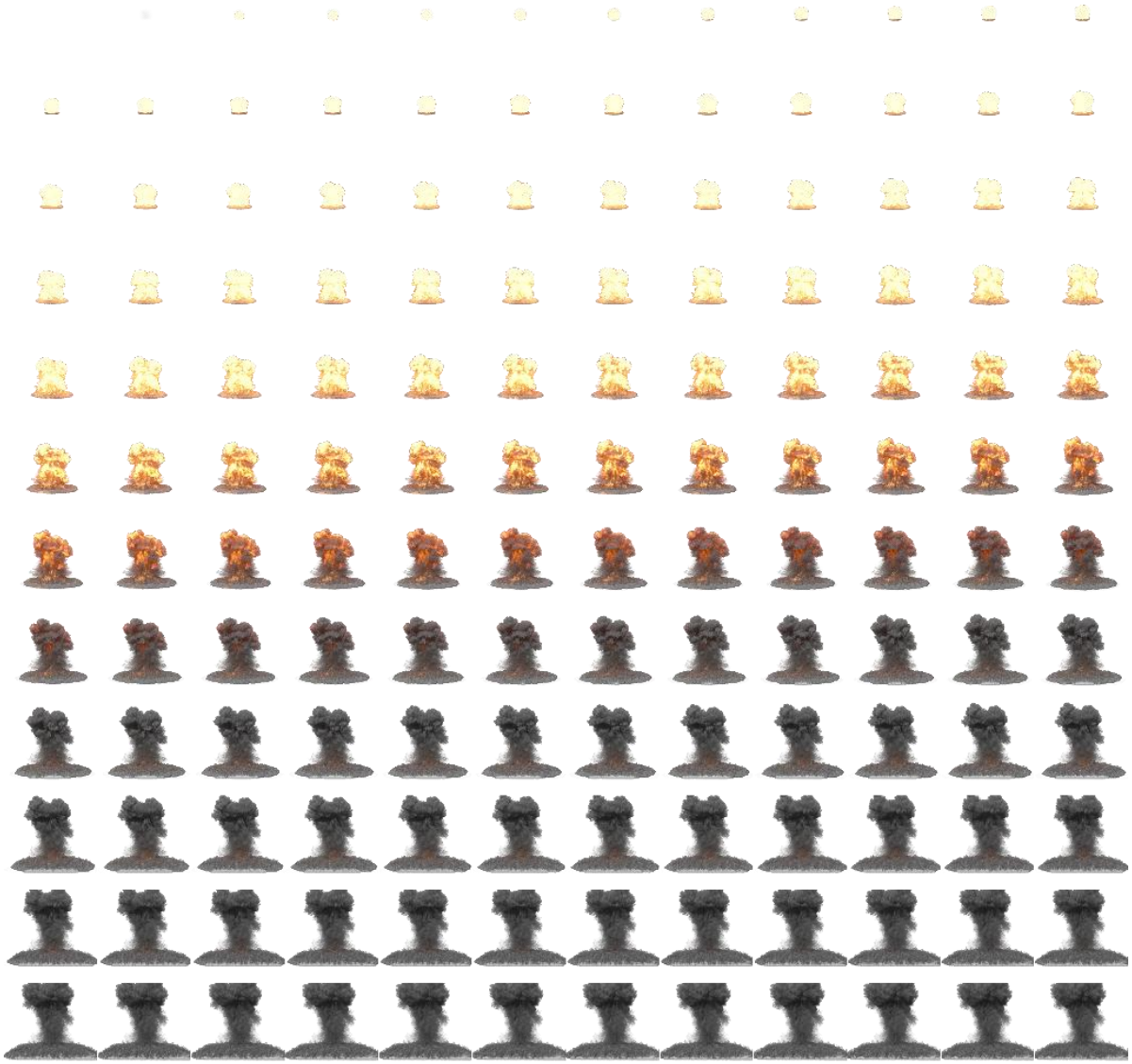


Concurrently, the project incorporates two custom materials to enhance the scene's realism. The first is the road surface material: by following online tutorials to blend source textures with a base color, while adjusting emissive and base color properties, adding roughness, and scaling dimensions, the project created a highway surface texture very similar to the real world. The second is the architectural material: by blending a beige base with a noise pattern, the project simulated marble veining, thereby ensuring that the building surfaces appear less pristine and sterile.





The greatest challenge in this project was balancing computer performance with the goal of rendering the nuclear mushroom cloud as three-dimensional and realistic as possible. Due to limited computing resources, the project did not utilize the VDB export method demonstrated in the tutorial. Instead, the project was exported as a sequence of PNGs and displayed on screen using a "flipbook" technique, like rapidly flipping 30 pages of a book every second. The consequence of this approach was that, while performance was significantly boosted, the mushroom cloud appeared as a 2D sprite in Unreal Engine and looked extremely blurry up close.



Furthermore, for the final showcase, the flipbook method proved incapable of triggering playback at a specific frame. At least, no solution to achieve this was discovered within the scope of this project. A workaround was adopted in which the mushroom cloud was set to loop continuously, allowing the desired visual effect to be achieved through subsequent post-production video editing.

Although the final output did not quite meet my expectations at the beginning of the semester, I remain reasonably satisfied with the mushroom cloud's rendering and the animation's camerawork. If I have the opportunity to continue refining this project, I hope to add more detail to the scene and make the mushroom cloud appear even more realistic.